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Our Common Future
Chapter 9: The Urban Challenge

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I. The Growth of Cities

1. The Crisis in Third World Cities

2. The Situation in Industrial World Cities

II. The Urban Challenge in Developing Countries

1. National Urban Strategies

2. Strengthening Local Authorities

3. Self-Reliance and Citizen Involvement

4. Housing and Services for the Poor

5. Tapping More Resources

III. International Cooperation

1. Cooperation Among Developing Countries

2. International Support

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1. By the turn of the century, almost half the world will live in urban areas - from small towns to huge megacities./1 The world's economic system is increasingly an urban one, with overlapping networks of communications, production, and trade./2 This system, with its flows of information, energy, capital, commerce, and people, provides the backbone for national development. A city's prospects - or a town's - depend critically on its place within the urban system, national and international. So does the fate of the hinterland, with its agriculture, forestry, and mining, on which the urban system depends.

2. In many nations, certain kinds of industries and service enterprises are now being developed in rural areas. But they receive high-quality infrastructure and services, with advanced telecommunications systems ensuring that their activities are part of the national (and global) urban-industrial system. In effect, the countryside is being 'urbanized'.

Table 9-1

Proportion of Population Living In Urban Areas, 1950-2000

	1950	1985	2000
	(Per cent)		
World Total	29.2	41.0	46.6
More Developed Regions	53.8	71.5	74.4
Less Developed Regions	17.0	31.2	39.3
Africa	15.7	29.7	39.0
Latin America	41.0	69.0	76.8
(Temperate South America)	(64.8)	(84.3)	88.6
(Tropical South America)	(35.9)	(70.4)	(79.4)
Asia	16.4	28.1	35.0
(China)	(11.0)	(20.6)	(25.1)
(India)	(17.3)	(25.5)	(34.2)
	(in Millions)		
World Total	734.2	1982.8	2853.6
More Developed Regions	447.3	838.8	949.9
Less Developed Regions		1144.0	1903.7
Africa	35.2	164.5	340.0
Latin America	67.6	279.3	419.7
Asia	225.8	791.1	1242.4

Source: 'Urban and Rural Population Projections. 1984 Unofficial Assessment', Population Division. United Nations, New York.

I. The Growth of Cities

3. This is the century of the 'urban revolution'. In the 35 years since 1950, the number of people living in cities almost tripled, increasing by 1.25 billion. In the more developed regions, the urban population nearly doubled, from 447 million to 838 million. In the less developed world, it quadrupled, growing from 286 million to 1.14 billion. (See Table 9-1.)

4. Over only 60 years, the developing world's urban population increased tenfold, from around 100 million in 1920 to close to 1 billion in 1980. At the same time, its rural population more than doubled.

- In 1940, only one person in eight lived in an urban centre, while about one in 100 lived in a city with a million or more inhabitants (a 'million city').

- By 1960, more than one in five persons lived in an urban centre, and one in 16 in a 'million city'.

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- By 1980, nearly one in three persons was an urban dweller and one in 10 a 'million city' resident./3

5. The population of many of sub-Saharan Africa's larger cities increased more than sevenfold between 1950 and 1980 Nairobi, Dar es Salaam, Nouakchott, Lusaka, Lagos, and Kinshasa among them./4 (See Table 9-2.) During these same 30 years, populations in many Asian and Latin American cities (such as Seoul, Baghdad, Dhaka, Amman, Bombay, Jakarta, Mexico City, Manila, Sao Paulo, Bogota, and Managua) tripled or quadrupled. In such cities, net immigration has usually been a greater contributor than natural increase to the population growth of recent decades.

6. In many developing countries, cities have thus grown far beyond anything imagined only a few decades ago and at speeds without historic precedent. (See Box 9-1.) But some experts doubt that developing nations will urbanize as rapidly in the future as in the last 10-40 years, or that megacities will grow as large as UN projections suggest. Their argument is that many of the most powerful stimuli to rapid urbanization in the past have less influence today, and that changing government policies could reduce the comparative attractiveness of cities, especially the largest cities, and slow rates or urbanization.

7. The urban population growth rate in developing countries as a whole has been slowing down from 5.2 per cent per annum in the late 1950s to 3.4 per cent in the 1950s./5 It is expected to decline even further in the coming decades. Nevertheless, if current trends hold. Third World cities could add a further three-quarters of a billion people by the year 2000. Over the same time, the cities of the industrial world would grow by a further 111 million./6

8. These projections put the urban challenge firmly in the developing countries, in the space of just 15 years (or about 5,500 days), the developing world will have to increase by 65 per cent its capacity to produce and manage its urban infrastructure, services, and shelter - merely to maintain present conditions. And in many countries, this must be accomplished under conditions of great economic hardship and uncertainty, with resources diminishing relative to needs and rising expectations.

Table 9-2

Examples of Rapid Population Growth in Third World Cities

	1950		Most Recent Figure		UN Projections for 2000
Mexico City	3.05		16.0	(1982)	22.3
Sao Paulo	2.7		12.6	(1980)	24.0
Bombay	3.0	(1951)	8.2	(1981)	16.0
Jakarta	1.45		6.2	(1977)	12.8
Cairo	2.5		8.5	(1979)	13.2
Delhi	1.4	(1951)	5.8	(1981)	9.6
Manila	1.78		5.5	(1980)	11.1
Lagos	0.27	(1952)	4.0	(1980)	8.3
Bogota	0.61		3.9	(1985)	9.6
Nairobi	0.14		0.83	(1979)	5.3
Dar es Salaam	0.15	(1960)	0.9	(1981)	4.6
Greater Khartoum	0.18		1.05	(1978)	4.3
Amman	0.03		0.78	(1978)	1.5
Nouakchott	0.0058		0.25	(1982)	1.1
Managua	0.11		0.51	(1980)	1.1
Santa Cruz	0.059		0.26	(1976)	1.0

Source: Recent census data used whenever possible; if none available, an estimate by the city government or a local research group has been used. UN projections for the year 2000 from Department of International Economic and Social Affairs, Estimates and Projections of Urban. Rural and City Populations 1950-2025 (the 1982 Assessment). ST/ESA/SER.R/58. New York. 1985 and from UN, Urban, Rural and City Populations 1950-

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2000 (as Assessed in 1978). Population Studies No. 68 (New York, 1980). Other data from Jorge E. Hardoy and David Satterthwaite, Shelter: Need and Response (Chichester, UK: John Wiley & Sons, 1981), with some figures updated with more recent census data.

1. The Crisis in Third World Cities

9. Few city governments in the developing world have the power, resources, and trained staff to provide their rapidly growing populations with the land, services, and facilities needed for an adequate human life: clean water, sanitation, schools, and transport. The result is mushrooming illegal settlements with primitive facilities, increased overcrowding, and rampant disease linked to an unhealthy environment.

Box 9-1

Dominating Cities

Nairobi, Kenya: In 1975, Nairobi had 57 per cent of all Kenya's manufacturing employment and two-thirds of its industrial plants. In 1979, Nairobi contained around 5 per cent of the national population.

Manila, Philippines: Metropolitan Manila produces one-third of the nation's gross national product, handles 70 per cent of all imports, and contains 60 per cent of the manufacturing establishments. In 1981, it contained around 13 per cent of the national population.

Lima, Peru: The metropolitan area of Lima accounts for 43 per cent of gross domestic product, for four-fifths of bank credit and consumer goods production, and for more than nine-tenths of capital goods production in Peru. In 1981, it was home to around 21 per cent of Peruvians.

Lagos, Nigeria: In 1978, Lagos' metropolitan area handled over 40 per cent of the nation's external trade, accounted for over 57 per cent of total value added in manufacturing, and contained over 40 per cent of Nigeria's highly skilled workers. It contains only some 5 per cent of the national population.

Mexico City, Mexico: In 1970, with some 24 per cent of Mexicans living there, the capital contained 30 per cent of the manufacturing jobs, 28 per cent of employment in commerce, 38 per cent of jobs in services, 69 per cent of employment in national government, 62 per cent of national investment in higher education, and 80 per cent of research activities. In 1965, it contained 44 per cent of national bank deposits and 61 per cent of national credits.

Sao Paulo, Brazil: Greater Sao Paulo, with around one-tenth of Brazil's national population in 1980, contributed one-quarter of the net national product and over 40 per cent of Brazil's industrial value-added.

Source: J.E. Hardoy and D. Satterthwaite, 'Shelter, Infrastructure and Services in Third World Cities', Habitat International, Vol. 10, No 4, 1986.

10. In most Third World cities, the enormous pressure for shelter and services has frayed the urban fabric. Much of the housing used by the poor is decrepit. Civic buildings are frequently in a state of disrepair and advanced decay. So too is the essential infrastructure of the city; public transport is overcrowded and overused, as are roads, buses and trains, transport stations, public latrines, and washing points. Water supply systems leak, and the resulting low water pressure allows sewage to seep into drinking water. A large proportion of the city's population often has no piped water, storm drainage, or roads./7

Given the distribution of incomes, given the foreseeable availability of resources national, local, and worldwide given present technology, and given the present weakness of local government and the lack of interest of national governments in settlement problems, I don't see any solution for the Third World city.

Third World cities are and they will increasingly become centres of competition for a plot to be invaded where you can build a shelter, for a room to rent, for a bed in a hospital, for a seat in a school or in a bus, essentially for the fewer stable adequately paid jobs, even for the space in a square or on a sidewalk where you can display and sell your merchandise, on which so many households depend.

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The people themselves organize and help construct most new housing units in Third World cities and they do so without the assistance from architects, planners, and engineers, nor from local or national governments. Furthermore, in many cases, national and local governments are frequently harassing these groups. The people themselves are becoming increasingly the true builders and designers of Third World cities and quite often the managers of their own districts.

Jorge Hardoy
International Institute for Environment and Development
WCED Public Hearing
Sao Paulo, 28-29 Oct 1985

11. A growing number of the urban poor suffer from a high incidence of diseases; most are environmentally based and could be prevented or dramatically reduced through relatively small investments. (See Box 9-2.) Acute respiratory diseases, tuberculosis, intestinal parasites, and diseases linked to poor sanitation and contaminated drinking water (diarrhoea, dysentery, hepatitis, and typhoid) are usually endemic; they are one of the major causes of illness and death, especially among children. In parts of many cities, poor people can expect to see one in four of their children die of serious malnutrition before the age of five, or one adult in two suffering intestinal worms or serious respiratory infections./8

12. Air and water pollution might be assumed to be less pressing in Third World cities because of lower levels of industrial development. But in fact hundreds of such cities have high concentrations of industry. Air, water, noise, and solid waste pollution problems have increased rapidly and can have dramatic impacts on the life and health of city inhabitants, on their economy, and on jobs. Even in a relatively small city, just one or two factories dumping wastes into the only nearby river can contaminate everyone's drinking, washing, and cooking water. Many slums and shanties crowd close to hazardous industries, as this is land no one else wants. This proximity has magnified the risks for the poor, a fact demonstrated by great loss of life and human suffering in various recent industrial accidents.

Box 9-2

Environmental Problems in Third World Cities

Out of India's 3,119 towns and cities, only 209 had partial and only 8 had full sewage and sewage treatment facilities. On the river Ganges, 114 cities each with 50,000 or more inhabitants dump untreated sewage into the river every day. DDT factories, tanneries, paper and pulp mills, petrochemical and fertilizer complexes, rubber factories, and a host of others use the river to get rid of their wastes. The Hoogly estuary (near Calcutta) is choked with untreated industrial wastes from more than 150 major factories around Calcutta. Sixty per cent of Calcutta's population suffer from pneumonia, bronchitis, and other respiratory diseases related to air pollution.

Chinese industries, most of which use coal in outdated furnaces and boilers, are concentrated around 20 cities and ensure a high level of air pollution. Lung cancer mortality in Chinese cities is four to seven times higher than in the nation as a whole, and the difference is largely attributable to heavy air pollution.

In Malaysia, the highly urbanized Klang Valley (which includes the capital, Kuala Lumpur) has two to three times the pollution levels of major cities in the United States, and the Klang river system is heavily contaminated with agricultural and industrial effluents and sewage.

Sources: Centre for science and Environment, State of India's Environment: A Citizens' Report (New Delhi: 1983); Vaclav Smil, The Bad Earth: Environmental Degradation in China (London: Zed Press, 1986); Sahabat Alam Malaysia, The State of Malaysian Environment 1983-84 - Towards Greater Environmental Awareness (Penang, Malaysia: 1983).

13. The uncontrolled physical expansion of cities has also had serious implications for the urban environment and economy. Uncontrolled development makes provision of housing, roads, water supply, sewers, and public services prohibitively expensive. Cities are often built on the most productive agricultural land, and unguided growth results in the unnecessary loss of this land. Such losses are most serious in nations with limited arable

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land, such as Egypt. Haphazard development also consumes land and natural landscapes needed for urban parks and recreation areas. Once an area is built up, it is both difficult and expensive to re-create open space.

14. In general, urban growth has often preceded the establishment of a solid, diversified economic base to support the build-up of housing, infrastructure, and employment. In many places, the problems are linked to inappropriate patterns of industrial development and the lack of coherence between strategies for agricultural and urban development. The link between national economies and international economic factors has been discussed in Part I of this report. The world economic crisis of the 1980s has not only reduced incomes, increased unemployment, and eliminated many social programmes, it has also exacerbated the already low priority given to urban problems, increasing the chronic shortfall in resources needed to build, maintain, and manage urban areas./9

2. The Situation in Industrial World Cities

15. The Commission's focus on the urban crisis in developing countries is not meant to imply that what transpires in the cities of the industrial world is not of crucial importance to sustainable development globally. It is. These cities account for a high share of the world's resource use, energy consumption, and environmental pollution. Many have a global reach and draw their resources and energy from distant lands, with enormous aggregate impacts on the ecosystems of those lands.

16. Nor is the emphasis on Third World cities meant to imply that problems within the cities of industrialized countries are not serious. They are. Many face problems of deteriorating infrastructure, environmental degradation, inner-city decay, and neighbourhood collapse. The unemployed, the elderly, and racial and ethnic minorities can remain trapped in a downward spiral of degradation and poverty, as job opportunities and the younger and better-educated individuals leave declining neighbourhoods. City or municipal governments often face a legacy of poorly designed and maintained public housing estates, mounting costs, and declining tax bases.

17. But most industrial countries have the means and resources to tackle inner-city decay and linked economic decline. Indeed, many have succeeded in reversing these trends through enlightened policies, cooperation between the public and private sectors, and significant investments in personnel, institutions, and technological innovation./10 Local authorities usually have the political power and credibility to take initiatives and to assess and deploy resources in innovative ways reflecting unique local conditions. This gives them a capacity to manage, control, experiment, and lead urban development. In centrally planned economies, the ability to plan and implement plans for urban development has been significant. The priority given to collective goods over private consumption may also have increased the resources available for urban development.

18. The physical environment in many cities of the industrial world has improved substantially over the decades. According to the historical records of many major centres - like London, Paris, Chicago, Moscow, and Melbourne - it was not too long ago that a major part of their population lived in desperate circumstances amid gross pollution. Conditions have improved steadily during the past century, and this trend continues, although the pace varies between and within cities.

Large cities by definition are centralized, manmade environments that depend mainly on food, water, energy, and other goods from outside. Smaller cities, by contrast, can be the heart of community-based development and provide services to the surrounding countryside.

Given the importance of cities, special efforts, and safeguards are needed to ensure that the resources they demand are produced sustainably and that urban dwellers participate in decisions affecting their lives. Residential areas are likely to be more habitable if they are governed as individual neighbourhoods with direct local participation. To the extent that energy and other needs can be met on a local basis, both the city and surrounding areas will be better off.

'Sustainable Development and How to Achieve It'
Global Tomorrow Coalition
WCED Public Hearing
Ottawa, 26-27 May 1986

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19. In most urban areas, almost everyone is served by refuse collection today. Air quality has generally improved, with a decline in the emission of particles and sulphur oxides. Efforts to restore water quality have met with a mixed record of success because of pollution from outside of cities, notably nitrates and other fertilizers and pesticides. Many coastal areas, however, close to major sewage outlets, show considerable deterioration. There is rising concern about chemical pollutants in drinking water and about the impacts of toxic wastes on groundwater quality. And noise pollution has tended to increase.

20. Motor vehicles greatly influence environmental conditions in the cities of the industrial world. A recent slowdown in the growth rate of vehicle numbers, stricter emission standards for new vehicles, the distribution of lead-free gasoline, improvements in fuel efficiency, improved traffic management policies, and landscaping have all helped reduce the impacts of urban traffic.

21. Public opinion has played a critical role in the drive to improve urban conditions. In some cities, public pressure has triggered the abandonment of massive urban development projects, fostered residential schemes on a more human scale, countered indiscriminate demolition of existing buildings and historic districts, modified proposed urban highway construction, and led to transformation of derelict plots into playgrounds.

22. The problems that remain are serious but they affect relatively limited areas, which makes them much more tractable than those of Cairo or Mexico City, for example. Certain aspects of urban decline even provide opportunities for environmental enhancement. The exodus of population and economic activities, while creating severe economic and social difficulties, reduces urban congestion, allows new uses for abandoned buildings, protects historic urban districts from the threat of speculative demolition and reconstruction, and contributes to urban renewal. The de-industrialization of these cities is often counterbalanced by the growth of the services sector, which brings its own problems. But this trend creates opportunities to remove heavy industrial pollution sources from residential and commercial areas.

23. The combination of advanced technology, stronger national economies, and a developed institutional infrastructure give resilience and the potential for continuing recovery to cities in the industrial world. With flexibility, space for manoeuvre, and innovation by local leadership, the issue for industrial countries is ultimately one of political and social choice. Developing countries are not in the same situation. They have a major urban crisis on their hands.

II. The Urban Challenge in Developing Countries

24. Settlements - the urban network of cities, towns, and villages - encompass all aspects of the environment within which societies' economic and social interactions take place. Internationally, the major cities of the world constitute a network for the allocation of investment and for the production and sale of most goods and services. These centres are the first to be plugged into this network, through air- and seaports and telecommunications. New technologies usually arrive and are first put into practice in large and then smaller cities. Only if centres are firmly connected to this network can they hope to attract investment in technologies and manufacturing goods for world markets. Nationally, cities are veritable incubators of economic activities. Some enterprises are large-scale but the vast majority are small, doing everything from selling snack foods to mending shoes and building houses. The growth of these activities is the foundation of the domestic economy.

1. National Urban Strategies

25. The natural evolution of this network of settlements, however, has caused apprehension in most developing countries. Of particular concern has been the phenomenal growth of often one or two major cities. In some countries, the desire to limit this growth has led to spatial policies designed to accelerate the development of secondary centres. Underlying this has been a particular concern that unbalanced growth is increasing interregional disparities and creating economic and social imbalances that can have serious consequences in terms of national unity and political stability.

26. Although far from conclusive, the available evidence suggests that most attempts by central governments to balance spatial development have been both expensive and ineffective. Major macroeconomic, social, and sectoral policies have often been directly opposed to the decentralization policy. Investments supported by governments and aid agencies have followed the same centralizing logic as private investments, and have built transportation facilities, educational and health institutions, and urban infrastructure and services where the

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demand exists - in the major city. Rural-urban migration has followed the same pattern. A major reason why so many migrants in recent decades went to cities such as Nairobi, Manila, Lagos, Mexico City, Sao Paulo, Rangoon, or Port au Prince was the dominant role each centre came to play in its national economy.

> We see that the increasing urban drift is inevitable: There are a lot of 'push' factors working in the rural areas. Rural pluralization is caused by absence of land reform, by the increase of absentee landownership, by the displacement of the Green Revolution.
>
> Resides the 'push' factors of the rural areas, there are, of course, the 'pull' factors, the glamour of the Big city, the higher pay of urban jobs as compared to rural income possibilities. So the informal sector of Jakarta has grown; maybe from the 7 million population of Jakarta, 3 or 4 million - at least two-thirds - are the result of the urban drift.
>
> George Adicondro
> Director, Irian Jaya, Rural Community Development Foundation
> WCED Public Hearing
> Jakarta, 26 March 1985

27. The macroeconomic and pricing policies pursued by governments further reinforced this concentration. The major cities, often the capital, usually receive a disproportionately large share of the total national expenditure on education and on subsidies to reduce the prices of water, corn, electric power, diesel fuel, and public transport. Railroad freight rates sometimes favour routes that pass through the capital. Property taxes in the city and surrounding districts may be undervalued. New or expanded industries given a boost by the import substitution policies are encouraged to establish in or near the capital./11

28. Agricultural and food policies have also tended to promote rapid growth of larger cities. Low or even negative economic supports for agricultural products have driven smallholders off their land and added to the numbers of the rural poor. Urban food prices, held low by subsidies, have served to attract many of them to cities. In recent years, however, some developing countries have found it possible to begin to shift more income from the major cities to the rural areas and smaller towns. In some cases, policies to promote small landholdings and intensive farming have had this effect. Increasing production, a growth in agricultural employment, and higher average incomes have stimulated the development of small and intermediate centres in the agricultural regions they serve./12

29. There are some important lessons to be learned about spatial strategies for urban development:

- Nothing much short of coercion will prevent the growth of the major city in the early stages of development.

- The key to successful intervention is timing, to encourage deconcentration only when the advantages of concentration are diminishing.

- Avoid policy interventions that increase the attractiveness of the major city, particularly subsidies on food and energy, overly generous provision of urban infrastructure and other services, and excessive concentration of administrative power in the capital.

- The best way to encourage the growth of secondary centres is to build on the natural economic advantages of their regions, especially in resource processing and marketing, and the decentralized provision of government services.

- Rural and urban development strategies and approaches should be complementary rather than contradictory: The development of secondary centres is to the direct economic benefit of the resource areas they serve.

30. The job opportunities and housing provided by cities are essential to absorb the population growth that the countryside cannot cope with; as long as price controls and subsidies do not interfere, the urban market should offer advantages to rural producers. But there are obviously conflicts of interest between developing country city-dwellers and farmers. A major thrust of the discussion on food security (see Chapter 5) was to assert the importance of decisively turning the 'terms of trade' in favour of farmers, especially small farmers, through

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pricing and exchange rate policies. Many developing countries are not implementing such policies, partly for fear of losing the support of politically powerful urban factions. Thus they fail both to stem urban drift and to improve food security.

31. These considerations can provide the basis for developing an explicit national settlements strategy and policies within which innovative and effective local solutions to urban problems can evolve and flourish. Every government has such a strategy in effect, but it is most often implicit in a range of macroeconomic, fiscal, budget, energy, and agricultural policies. These policies have usually evolved incrementally in response to the pressures of the day and, almost invariably, they contradict each other and the stated settlement goals of the government. A national urban strategy could provide an explicit set of goals and priorities for the development of a nation's urban system and the large, intermediate, and small centres within it. Such a strategy must go beyond physical or spatial planning, it requires that governments take a much broader view of urban policy than has been traditional.

32. With an explicit strategy, nations can begin to reorient those central economic and major sectoral policies that now reinforce megacity growth, urban decline, and poverty. They can likewise promote more effectively the development of small and intermediate urban centres, the strengthening of their local governments, and the establishment of services and facilities needed to attract development initiatives and investment. Ministries of Planning, Finance, Industry, Agriculture, and so on would have clear goals and criteria against which to assess the effects of their policies and expenditures on urban development. Contradictory policies and programmes could be changed. At the very least, the spatial biases inherent in macroeconomic and fiscal policies, annual budgets, pricing structures, and sectoral investment plans could be exposed and assessed. Within such a strategy, the traditional tools of urban policy, including land use planning and control, would stand a better chance of being effective.

33. The formulation of such a strategy is clearly a central government responsibility. Beyond this, however, the role of central governments should be primarily to strengthen the capacity of local governments to find and carry through effective solutions to local urban problems and stimulate local opportunities.

2. Strengthening Local Authorities

34. The institutional and legal structures of local government in most developing nations are inadequate for these purposes. In most African and Asian nations the structure of urban government goes hack to the colonial period and was designed to deal with predominantly rural and agricultural societies. It was never intended to cope with rapid urbanization or to manage cities of several million inhabitants. Newly independent governments inherited a framework of laws and procedures totally inappropriate to deal with the urban processes they were about to confront. Yet in many nations, this inherited framework remains largely in place.

35. Where the immediate colonial past is less evident, as in most Latin American nations, the political, institutional, and legal frameworks for local government are often just as inappropriate and inadequate. As in Asia and Africa, most are based on models imported from Europe or North America. This has made it difficult for them to influence the direction of urbanization and to manage the problems of large, rapidly expanding urban centres. It has created cities that are energy and material-intensive and dependent on imports, and that add to the burden on the national economy, including pressures on trade and balance of payments.

36. Urban development cannot be based on standardized models, imported or indigenous. Development possibilities are particular to each city and must be assessed within the context of its own region. What works in one city may be totally inappropriate in another. Although technical help from central agencies may be needed, only a strong local government can ensure that the needs, customs, urban forms, social priorities, and environmental conditions of the local area are reflected in local plans for urban development. But local authorities have not been given the political power, decision making capacity, and access to revenues needed to carry out their functions. This leads to frustration, to continuing criticism of local government for insufficient and inefficient services, and to a downward spiral of weakness feeding on weakness.

> A lot of youth in the Third World countries and even adults are unemployed. We want simple technologies whereby one particular person can do a kind of a job that could have provided job opportunities to several hundreds. What are we doing with the surplus potential energy? So again I say that development is people, it is not high technology, it is not modernization, it is not westernization. But it should be culturally relevant.

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Jan Selego
World Vision International
WCED Public Hearing
Nairobi, 23 Sept 1986

37. The lack of political access is a major weakness of local government in many developing countries. Most local governments have difficulties getting enough revenue to cover their operating expenses, let alone to make new investments to extend services and facilities. Even richer city governments have access to the equivalent of only $10-50 per inhabitant to invest each year. Despite these weaknesses, the trend in recent decades has been for national governments to reduce the financial capacity of local governments in real terms.

38. The result is growing centralization and continuing weaknesses at both the central and local level. Instead of doing a few things well, central authorities end up doing too many things, none of them well. Human and financial resources get stretched too thin. Local governments do not gain the expertise, authority, and credibility needed to deal with local problems.

39. To become key agents of development, city governments need enhanced political, institutional, and financial capacity, notably access to more of the wealth generated in the city. Only in this way can cities adapt and deploy some of the vast array of tools available to address urban problems - tools such as land title registration, land use control, and tax sharing.

3. Self-Reliance and Citizen Involvement

40. In most developing countries between one-fourth and one-half of the economically active urban population cannot find adequate, stable livelihoods. With few jobs available in established businesses or government services, people have to find or create their own sources of income. These efforts have resulted in the rapid growth of what has been termed the 'informal sector', which provides much of the cheap goods and services essential to city economies, business, and consumers.

41. Thus, while many poor people may not be officially employed, most are working - in unregistered factories end construction firms, selling goods on street corners, making clothes in their homes, or as servants or guards in better-off neighbourhoods. Most of the so-called unemployed are in fact working 10-15 hours a day, six to seven days a week. Their problem is not so much underemployment as underpayment.

42. Most house building, maintenance, or upgrading in the cities of developing countries is done outside official plans and usually in illegal settlements. This process mobilizes untapped resources, contributes to capital formation, and stimulates employment. These informal sector builders represent an important source of urban employment, in particular for low and unskilled labour. They are not capital- or technology-intensive, they are not energy-intensive, and as a rule they do not impose a drain on foreign exchange. In their way, they contribute their share to attaining some of the nation's major development objectives. Moreover, they are flexible in responding to local needs and demands, catering in particular to poorer households, which usually have nowhere else to turn. Many governments have begun to see the wisdom of tolerating rather than quashing their work. Large-scale bulldozing of squatter communities is now rarer, although it still happens.

43. Governments should give more support to the informal sector, recognizing its vital functions in urban development. Some governments have done so, facilitating loans and credit to small entrepreneurs, building cooperatives, and neighbourhood improvement associations. Providing tenure to those living in illegal settlements is basic to this process, as is easing some building and housing regulations.

44. Multilateral and bilateral development assistance agencies should follow suit, and some are beginning to do so. Non-governmental and private voluntary organizations are springing up in many countries to provide cost-effective channels for assistance, ensuring that it gets to those who can use it. A much larger proportion of assistance could be channelled directly through these organizations.

45. The above measures would also reinforce self-reliance and local governance by the poor in their own neighbourhood associations. Left to their own devices, the poor in many Third World cities have organized to fill gaps in services left by the local government. Among other things, community groups mobilize and organize

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fund-raising or mutual self-help to deal with security, environmental, and health problems within the immediate area.

46. Governments should move from a position of neutrality or antagonism to active support for such efforts. A few have actually institutionalized such programmes so that public ministries or agencies work continuously with community organizations. In the Indian city of Hyderabad, for example, an Urban Community Development Department set up by the municipal corporation works directly with community groups and non-government organizations in poorer neighbourhoods. By 1983, some 223 organizations had been formed by residents in low-income areas, plus 135 youth organizations and 99 women's groups./13 In this way governments can become partners and sponsors of the people who are the main builders of their cities.

> The shantytowns have found their own technique, their own resources without any assistance from anyone else, and they solved their housing problems. The real problem is not that. It is the poverty, the lack of planning, the lack of technical assistance, the lack of financing to buy construction materials, the lack of urban equipment.
>
> To change this housing policy for human settlements, they should stimulate self-construction, instead of financing these large housing complexes. It would have been much better and would have cost less to help the people to carry out the self-construction.
>
> Generally speaking, it seems clear that without meeting the basic needs of human beings, concern for the environment has to be secondary. Man has to survive, answer, and attend first to his basic survival needs - food, housing, sanitation - and then to the environment.
>
> Walter Pinto Costa
> President, Environmental and Sanitation Association
> WCED Public Hearing
> Sao Paulo, 28-29 Oct 1985

4. Housing and Services for the Poor

47. In most developing-world cities, there is little low-cost housing. Generally those on low incomes either rent rooms - whether in tenements or cheap boarding-houses, or in someone else's house or shack - or they build or buy a house or shack in an illegal settlement. There are many kinds and degrees of illegality, and these influence the extent to which governments tolerate the existence of such settlements, or even provide them with public services and facilities.

48. Whatever form it takes, low-income accommodation generally shares three characteristics. First, it has inadequate or no infrastructure and services - including piped water, sewers, or other means of hygienically disposing of human wastes. Second, people live in crowded and cramped conditions under which communicable diseases can flourish, particularly when malnutrition lowers resistance. Third, poor people usually build on land ill-suited for human habitation: floodplains, dusty deserts, hills subject to landslide, or next to polluting industries. They choose these sites because the land's low commercial value means they stand a better chance of not being evicted.

49. Landownership structures and the inability or unwillingness of governments to intervene in these structures are perhaps the main factors contributing to 'illegal' settlements and chaotic urban sprawl. When half or more of a city's workforce has no chance of obtaining a legal plot on which a house can be built, let alone of affording to buy or rent a house legally, the balance between private landownership rights and the public good must be quickly rethought.

50. Given urbanization trends in most developing countries, there is no time to wait for slow and uncertain programmes. Government intervention must be reoriented so that limited resources are put to maximum effect in improving housing conditions for the poor. The options for intervention are many (see Box 9-3), but governments should be guided by these seven priorities:

- provide legal tenure to those living in 'illegal' settlements, with secure titles and basic services provided by public authorities;

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- ensure that the land and other resources people need to build or improve their housing are available;
- supply existing and new housing areas with infrastructure and services;
- set up neighbourhood offices to provide advice and technical assistance on how housing can be built better and cheaper, and on how health and hygiene can be improved;
- plan and guide the city's physical expansion to anticipate and encompass needed land for new housing, agricultural land, parks, and children's play areas;
- consider how public intervention could improve conditions for tenants and those living in cheap rooming or boarding-houses; and
- change housing finance systems to make cheap loans available to lower-income and community groups.

51. Most cities urgently need a large and continuous increase in the availability of cheap housing plots convenient to the main centres of employment. Only government intervention can achieve this, but no general prescriptions are possible. Societies differ too much in how they view private landownership and land use rights, in how they use different instruments such as direct grants, tax write-offs, or deduction of mortgage interest, and in how they treat land speculation, corruption, and other undesirable activities that often accompany processes of this kind. Although the means are particular to each nation, the end must be the same: governments ensuring that there are cheaper, better-serviced, better-located, legal alternatives to illegal plots. If this need is not met, the uncontrolled growth of cities - and its accompanying high costs - will not be stopped.

52. Besides land, building materials are another major cost for people putting up their own houses. Government support for the production of materials and of certain structural components, fixtures, and fittings could reduce housing costs and create many jobs. Small neighbourhood workshops often have cost advantages because of the low cost of transport from the workshop to the building site.

Box 9-3

Three Ways to Use $20 Million to Improve Conditions in a City of 1 Million

Option 1:

Build 2,000 public housing units for poor families (with an average of six family members), each costing $10,000. Conditions are improved for 12,000 people, but little cost recovery is possible for poor families. If the city's population grows at 5 per cent annually, 630.000 new inhabitants will be added over 10 years, so only a tiny fraction of total population will have benefited.

Option 2:

Establish a 'site-and-service scheme', whereby poor families are responsible for building their houses on an allocated site supplied with piped water, connection to a sewer system, and electricity, roads, and drainage. At $2,000 per plot, this means housing for some 60,000 people - about 10 per cent of the city's population growth over 10 years.

Option 3:

Allocate $100,000 to a neighbourhood organization representing 1,000 poor households (6,000 people) in an existing low-income settlement. It chooses to improve drainage and roads, build a health clinic, establish a cooperative to produce inexpensive building materials and components, and reblock the settlement to improve access roads and provide 50 new plots. With $10 million, 100 such community initiatives are supported, reaching 600,000 people and providing 5,000 new housing plots. Many new jobs are stimulated. The remaining $10 million is spent on installing piped water; at $100 per household, all 600,000 people reached.

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53. The majority of building codes and standards are ignored because following them would produce buildings too expensive for most people. A more effective approach might be to set up neighbourhood offices to provide technical advice on how health and safety can be improved at minimum cost. Good professional advice can lower building costs and improve quality, and might be more effective than prescribing what can or cannot be built.

54. Many poor people rent accommodation; half or more of a city's entire population may be tenants. Increasing the availability of house sites, materials, and credits does little for those who must rent. One possibility is financial support to non-governmental, non-profit organizations to purchase end develop property specifically for rental units. A second is support for tenants to buy out landlords and convert tenancy into cooperative ownership.

55. Governments, especially those strapped for resources, may claim that piped water supplies and sewage disposal systems are too expensive. As a consequence, poor people may have to pay water vendors far more per litre of water than middle- or upper-income groups pay public agencies to pipe water into their homes. Western water-borne sewage systems and treatment plants may be prohibitively expensive. But other techniques and systems cost between one-tenth and one-twentieth as much per household, and most of these use much less water. Moreover, lower-cost technology can be upgraded over time, as money becomes available./14

56. Major improvements can be made relatively cheaply in all these areas. But costs will remain low only if low-income groups are encouraged to participate fully in defining what they need, in deciding what they will contribute to the new services, and in doing the job with their own hands. This cooperation depends on establishing the new relationship between citizens and government called for earlier.

5. Tapping More Resources

57. The available resources in or close to cities are often underused. Many landowners leave well-located sites undeveloped in order to benefit later from their increasing value as the city grows. Many public agencies have land that could be put to better use, such as the area next to stations and harbours controlled by railway and port authorities. Several countries have introduced special programmes to encourage public and private cooperation in the development of such lands, a trend that should be encouraged. There is a general need to find innovative and effective ways of pooling land for the common good. Most cities have mechanisms for acquiring land either at market rates (which means that schemes are never implemented), or at arbitrarily low confiscatory rates (where the alliance of political forces and landlords blocks the acquisition anyway).

58. Governments should also consider supporting urban agriculture. This may have less relevance in cities where land markets are highly commercialized and land for housing is in short supply. But in most cities, especially those with less commercialized land markets, considerable potential exists. Many African cities already realize this. Urban agriculture, especially on city fringes, is undertaken by people as a way to feed themselves, in other instances, the process is more commercialized, with enterprises specializing in vegetable production for sale within the city.

59. Officially sanctioned and promoted urban agriculture could become an important component of urban development and make more food available to the urban poor. The primary purposes of such promotion should be to improve the nutritional and health standards of the poor, help their family budgets (50-70 per cent of which is usually spent on food), enable them to earn some additional income, and provide employment. Urban agriculture can also provide fresher and cheaper produce, more green space, the clearing of garbage dumps, and recycling of household waste./15

> I'm an expert in slum dwelling. We're establishing a small, tiny organization trying to organize slum dwellers, because we see so many slums. Slums in the city, slums in the villages, slums in the forests.
>
> I have worked for four years to motivate my fellow slum dwellers to become transmigrants, and they finally migrated to ten places all over Indonesia. They are still in very good communication with me. They're still sending me letters, and they say that life is not better in the transmigration areas. Living in the shadows in the urban slums or living in the shadows in the transmigration site is just the same.

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When I go back to my people, the slum dwellers, tonight they will ask me what I have got from this meeting in the big hotel. They won't ask for information, just 'Have you brought some money for us to build new houses?'

Syamsuddin Nainggolan
Founder, Yayasan Panca Bakti
WCED Public Hearing
Jakarta, 26 March 1986

60. Another poorly used resource is solid wastes, the disposal of which has become a major problem in many cities, with much of it dumped and uncollected. Promoting the reclamation, reuse, or recycling of materials can reduce the problem of solid waste, stimulate employment, and result in savings of raw materials. Composting can support urban agriculture. If a municipal government lacks the resources to collect household wastes regularly, it can support existing community-based schemes. In many cities, literally thousands of people already make a living sorting through wastes by hand on municipal tips. Investing in a more capital-intensive, automatic recycling plant could be doubly counterproductive if it unnecessarily consumes scarce capital or if a plant would destroy many people's livelihoods. But an immediate need here is to give health advice and provide health care services to those who are making a living off municipal tips./16

III. International Cooperation

61. The future will be predominantly urban, and the most immediate environmental concerns of most people will be urban ones. The effectiveness of efforts to improve urban life depends largely on the health of national economies. In many developing countries, this is linked closely to the state of the world economy. An improvement in international economic relations (see Chapter 3) would perhaps do more than anything else to enhance the capacity of developing countries to address their linked urban and environmental problems. But beyond that is the need to strengthen cooperation among developing countries and to increase various types of direct support from the international community.

1. Cooperation Among Developing Countries

62. Developing countries can do a great deal together to develop the policy concepts, programmes, and institutions needed to tackle the urban crisis they share. Although the management problems confronting Caracas, Dakar, or Delhi have little relevance to those confronting London or Paris, the cities of Latin America, West Africa, or South Asia have much in common. As they formulate broad national urban strategies, it is important that they share experiences on the management of their growing megacities, on the development of small and intermediate centres, on strengthening local government, on upgrading illegal settlements, on crisis-response measures, and on a range of other problems that are more or less unique to the Third World.

63. Further research could provide the basis for rethinking the Third World city. It could also feed in-country training programmes (or, for smaller nations, regional training programmes) for city and municipal government staff. Good policy proposals and good training courses depend on good local information and analysis; far too little of all three of these is found within developing countries and cities.

2. International Support

64. A greater flow of international resources is required to support the efforts of developing countries to tackle the unfolding urban crisis. An agreed definition of 'urban development assistance' does not exist, but the Development Assistance Committee recently estimated that total bilateral and multilateral aid for urban programmes averaged about $900 million per year over 1980-84./17 It is also estimated that to date fewer than 5 per cent of the developing world's urban population has been reached by a housing or neighbourhood upgrading project sponsored by a development assistance agency. This level of support needs to be increased significantly. Moreover, the scope of support should be broadened and its quality and terms improved.

65. In addition, development assistance agencies should increase aid and technical assistance in three areas:

- to set up infrastructure funds for local governments;

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- to undertake tasks such as reorganizing local tax assessments and collection, preparing or updating maps of property ownership, and setting up technical teams to advise households and community groups on improving housing;

- for in-country training courses and on-the job training for local officials.

66. Part of the increased aid should go directly to community groups, using intermediaries such as national or international NGOs. Several bilateral aid programmes have already demonstrated the cost-effectiveness of this approach; various NGOs have been responsible for many successful community based schemes to improve housing and provide basic services. They are generally more successful at reaching the poorest. More aid should also go to supporting independent research groups working in housing and urban issues, particularly those providing advice to local governments and community groups; many are doing so already, especially in Latin America.

Box 9-4

Misunderstanding Women's Needs in Housing Projects

Housing projects often use a gridiron layout that does not allow women to work in their house and at the same time keep an eye on their own or their neighbours' children. House designs and plot sizes rarely consider the fact that many women will want to use their houses as workshops (to make clothes, for instance) or as shops, which in fact are often forbidden in low-income housing projects. Application procedure for low-income housing sometimes requires 'husbands' to apply; this excludes women-headed households - between 30 and 50 per cent of all households. Women's special needs in different cultures are ignored - in Islamic societies, for example, women's need for private open space within the house is rarely considered in house designs, while their need for relatively sheltered pathways to get to shops and clinics is not acknowledged in site layouts.

Source: Based on C.O.N. Moser, 'Housing Policy: Towards a Gender Awareness Approach', Working Paper #71, Development Planning Unit, London, 1985.

67. International cooperation can also contribute to developing low cost technologies for urban needs and studying ways of meeting the housing needs of women. (See Box 9-4.)

68. Many technical agencies within the UN system have the appropriate knowledge bases to play a valuable role in advising and supporting governments, notably the UN Centre for Human Settlements (UNCHS, or Habitat). They should identify the information and guidelines that city governments need and the form in which it can be made accessible and usable by them. This could be patterned, for example, upon the ongoing efforts to prepare guidebooks for community workers on identifying disease vectors and mobilizing communities to deal with them, and on interventions to promote child survival and health. More generally. Habitat can strengthen international cooperation at the global level, as in the UN International Year of Shelter for the Homeless. The capacity of the UN system to provide leadership on human settlements issues through Habitat needs to be strengthened.

Footnotes

1/ This chapter draws heavily on four background papers prepared for WCED: I. Burton, 'Urbanization and Development', 1985; J.E. Hardoy and D. Satterthwaite, 'Shelter, Infrastructure and Services in Third World Cities', 1985 (printed in Habitat International, Vol. 10, No. 4, 1986); J.E. Hardoy and D. Satterthwaite, 'Rethinking the Third World City, 1986; and I. Sachs, 'Human Settlements: Resource and Environmental Management', 1985.

2/ See J. Jacobs, Cities and the Wealth of Nations (New York: Random House, 1984).

3/ UN, The Growth in the World's Urban and Rural Population 1920-1980, Population Studies No. 44 (New York: 1969); UN, Urban, Rural and City Populations 1950-2000 (as assessed in 1978), Population Studies No. 68 (New York: 1980).

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4/ The expansion of 'city' or 'metropolitan area' boundaries accounts for some of the population growth in Table 9 2. The UN projections are based on extrapolating past trends. This method often provides a poor guide to future trends, especially long-term ones. But the data base with which to make better projections ie not available.

5/ UNCHS (Habitat) position paper for October 1986 DAC meeting on Urban Development, OECD document DAC (86)47, 27 August 1986.

6/ Department of International Economic and Social Affaire, 'Urban and Rural Population Projections, 1984' (unofficial assessment), UN, New York, 1986.

7/ J.E. Hardoy and D. Satterthwaite, Shelter: Need and Response; Housing, Land and Settlement Policies in Seventeen Third World Nations (Chichester, UK: John Wiley & Sons, 1981). For the situation in Sao Paulo, See Jorge Wilhelm, 'Sao Paulo: Environmental Problems of the Growing Metropolis', submitted to WCED Public Hearings, Sao Paulo, 1985.

8/ J.E. Hardoy and D. Satterthwaite, 'Third World Cities and the Environment of Poverty', Geoforum, Vol. 15, No. 3, 1984. See also World Social Prospects Association, The Urban Tragedy (Geneva: UNITAR, 1986).

9/ See Osvaldo Sunkel, 'Debt, Development and Environment', submitted to WCED Public Hearings, Sao Paulo, 1985; Ricardo Jordan S., 'Population and the Planning of Large Cities in Latin America', paper submitted to the International Conference on Population and the Urban Future, Barcelona, Spain, 19-22 May 1986.

10/ G. Scimemi, 'Cita e Ambiente', DAEST, Instituto Universtario di Architectura, Venezia, 1987. See also, The State of the Environment in OECD Member Countries (Paris: OECD, 1979 and 1985).

11/ I. Scott, Urban and Spatial Development in Mexico (London: Johns Hopkins University Press, 1982).

12/ See Chapter 8 in J.E. Hardoy and D. Satterthwaite (eds.), Small and Intermediate Urban Centres; Their role in Regional and National Development in the Third World (London: Hodder and Stoughton, 1986).

13/ UNCHS, 'Habitat Hyderabad Squatter Settlement Upgrading Project, India', project monograph produced for the International Year of Shelter for the Homeless, Nairobi, 1986

14/ J. M. Kalbermatten et al., Appropriate Technology for Water Supply and Sanitation; a Summary of Technical and Economic Options (Washington DC: World Bank, 1980).

15/ D. Silk, 'Urban Agriculture', prepared for WCED, 1986.

16/ N. Khouri-Dagher, 'Waste Recycling: Towards Greater urban Self-Reliance', prepared for WCED, 1985.

17/ See draft annotated agenda for October 1986, DAC Meeting on Urban Development. OECD document DAC (86)15. The World Bank definition of urban development assistance was used, which includes fostering urban efficiency and alleviating poverty, shelter, urban transport, integrated urban development, and regional development on secondary cities.

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